STPM 2013 PHYSICS 960/2 (U2)

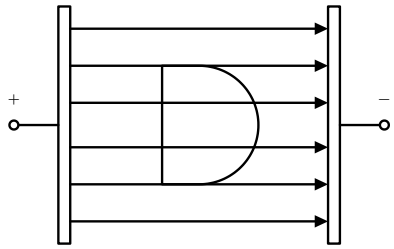
Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $g = 9.81$ m s^{-2}, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}.

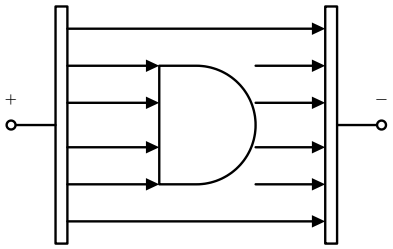
Section A [15 marks]

1. Which diagram shows the resultant electric field when a conducting plate is placed in between two parallel metal plates which are connected to a constant voltage supply?

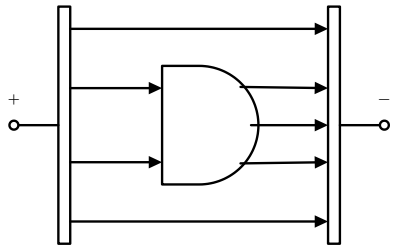
A



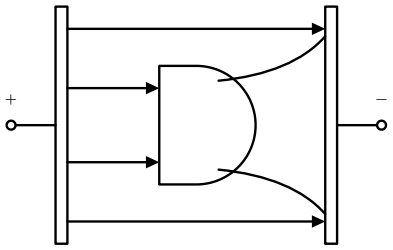
B



C



D



Answer: C

Field lines end on the induced negative charges on the left face of the conducting plate and start again from the induced positive charges on its right face, meeting the surfaces at right angles; there is no field inside the conductor (C).

Explanation: Inside a conductor in electrostatic equilibrium $E = 0$.

2. Three positive point charges each of magnitude Q are placed one after another at each corner of an equilateral triangle of side a. What is the total potential energy of the system of charges?

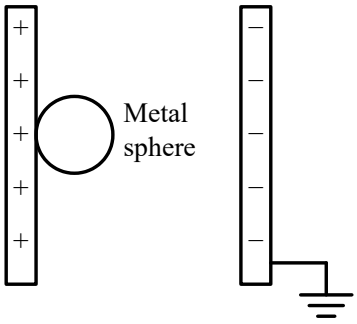
A $\frac{3Q}{4\pi\varepsilon_0 a}$ B $\frac{3Q^2}{4\pi\varepsilon_0 a}$ C $\frac{3Q}{4\pi\varepsilon_0 a^2}$ D $\frac{3Q^2}{4\pi\varepsilon_0 a^2}$

Answer: B

Three pairs, each with $U = \frac{Q^2}{4\pi\varepsilon_0 a}$: total $U = \frac{3Q^2}{4\pi\varepsilon_0 a}$

Explanation: Count each pair of charges once.

3. An air-filled parallel plate capacitor of capacitance C_1 which is charged to a potential difference of V is shown in the diagram below.



An isolated neutral metal sphere of capacitance C_2 is brought into contact with the positive plate. What is the charge gained by the metal sphere?

A $\frac{1}{2}C_1V$ B $\frac{C_1C_2}{C_1+C_2}V$ C $\frac{C_1^2}{C_1+C_2}V$ D $\frac{C_2^2}{C_1+C_2}V$

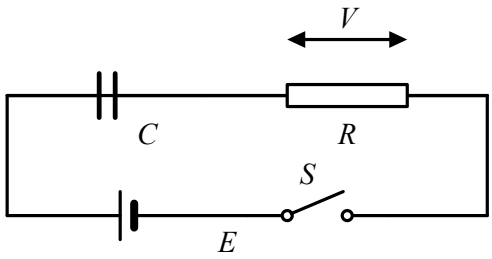
Answer: B

The charge C_1V on the positive plate is shared until the plate and the sphere are at the same potential:

$$\frac{q}{C_2} = \frac{C_1V - q}{C_1} \Rightarrow q = \frac{C_1C_2}{C_1+C_2}V$$

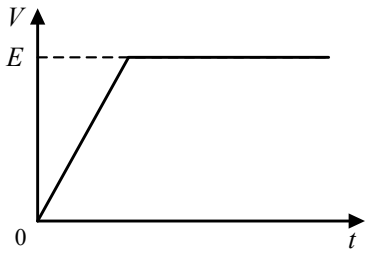
Explanation: Charge is conserved and shared in proportion to capacitance.

4. An uncharged capacitor C which is connected in series with a resistor R and a battery of e.m.f. E is shown in the circuit below.

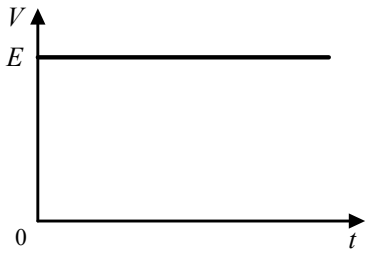


Which graph represents the variation of the potential difference V across the resistor with time t after the switch S is closed?

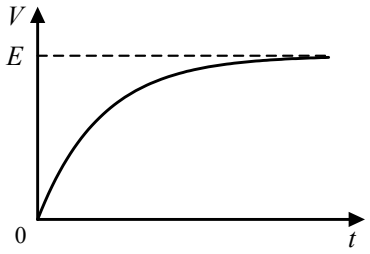
A



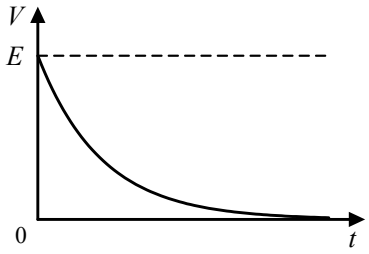
B



C



D

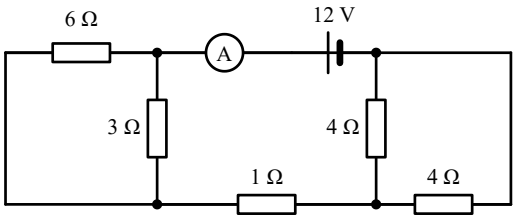


Answer: D

At first the capacitor is uncharged, so all of E is across R; as the capacitor charges, $V_R = Ee^{-t/RC}$ decays to zero (D).

Explanation: $V_R + V_C = E$ at all times.

5. A network of resistors connected to an ammeter and a cell of e.m.f. 12 V of negligible internal resistance is shown in the circuit below.



What is the reading of the ammeter in the circuit?

A 6.0 A B 4.0 A C 2.4 A D 1.5 A

Answer: C

The left wire joins the top-left and bottom-left; the right wire joins the top-right and bottom-right. 6 Ω ‖ 3 Ω = 2 Ω; the two 4 Ω resistors are in parallel = 2 Ω; with the 1 Ω in series: $R = 2 + 1 + 2 = 5\ \Omega$

$$I = \frac{12}{5} = 2.4\ \text{A}$$

Explanation: Redraw the circuit by labelling the points joined by plain wires as the same node.

6. The magnitude of the drift velocity of the charge carriers in a semiconductor is dependent on the following **except**

A the type of charge
B the electric current
C the cross-sectional area
D the number of charge per unit volume

Answer: A

$v = \frac{I}{nAq}$: the magnitude depends on I, A, n and the size of the charge, but not on its sign (type) (A)

Explanation: The type of charge decides only the direction of drift.

7. A battery of e.m.f. 12.0 V and internal resistance 0.050 Ω is connected to a 2.00 Ω resistor. What are the potential difference across the battery and the power dissipated by the internal resistance of the battery?

	Potential difference	*Power dissipated*
A	11.7 V	1.70 W
B	11.7 V	68.5 W
C	12.0 V	1.70 W
D	12.0 V	70.2 W

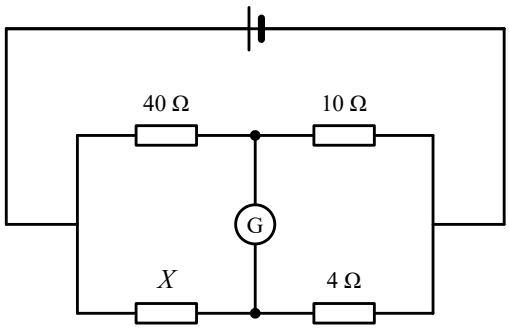
Answer: A

$$I = \frac{12.0}{2.00 + 0.050} = 5.85 \text{ A}; \; V = IR = (5.85)(2.00) = 11.7 \text{ V}$$

$$P = I^2 r = (5.85)^2(0.050) = 1.71 \text{ W}$$

Explanation: The power delivered to the 2.00 Ω resistor is 68.5 W.

8. A circuit consisting of four resistors and a galvanometer which is connected to a battery is shown in the diagram below.



If the reading of the galvanometer is zero, what is the resistance of X?

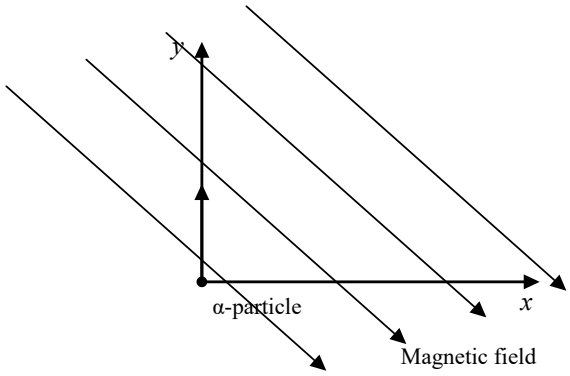
A 1 Ω B 16 Ω C 46 Ω D 100 Ω

Answer: B

$$\text{Balanced bridge: } \frac{40}{10} = \frac{X}{4} \Rightarrow X = 16 \text{ Ω}$$

Explanation: The ratio of resistances is the same on both sides of the galvanometer.

9. An α-particle which is moving in a positive-*y* direction in a region of uniform magnetic field is shown in the diagram below.



What is the direction of the magnetic force acting on the α-particle?

A Positive-*x* direction
B Negative-*x* direction
C Out of the paper
D Into the paper

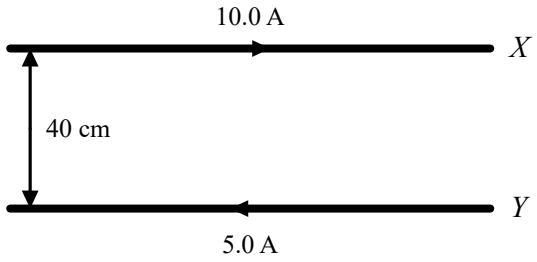
Answer: D

F = *q***v** × **B**: **v** along +*y* and **B** in the *xy*-plane pointing along +*x* and −*y*; only the +*x* component of **B** gives a force: $+y \times +x = -z$

The α-particle is positive, so the force is into the paper.

Explanation: The component of **B** parallel to **v** produces no force.

10. Two long parallel wires *X* and *Y* which are separated by a distance of 40 cm is shown in the diagram below.



The currents in *X* and *Y* which are in opposite directions are 10.0 A and 5.0 A respectively. The force per unit length between the two wires is

A attractive force with a magnitude of 2.5×10^{-5} N m^{-1}
B repulsive force with a magnitude of 2.5×10^{-5} N m^{-1}
C attractive force with a magnitude of 5.0×10^{-5} N m^{-1}
D repulsive force with a magnitude of 5.0×10^{-5} N m^{-1}

Answer: B

$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d} = \frac{(4\pi \times 10^{-7})(10.0)(5.0)}{2\pi(0.40)} = 2.5 \times 10^{-5} \text{ N m}^{-1}$$

Opposite currents repel.

Explanation: Each wire feels the same force (Newton's third law).

11. As blood is an electrically conducting fluid, the velocity of blood in an artery could be measured electromagnetically. This is done by measuring the potential difference produced across the artery when a magnetic field is set up across it. The method used to measure the velocity of blood is based on

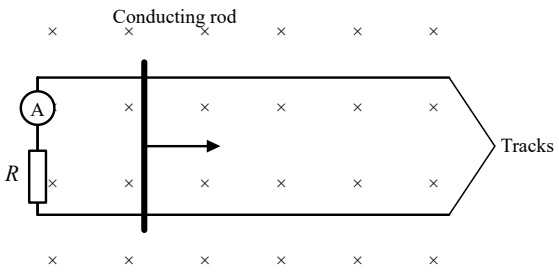
A the Hall effect
B the Doppler effect
C the photoelectric effect
D the thermoelectric effect

Answer: A

Ions in the moving blood are deflected sideways by the magnetic field, producing a p.d. across the artery: the Hall effect (A).

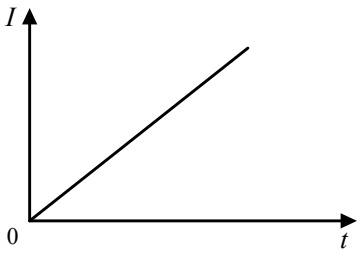
Explanation: $V = Bvd$, so v can be found.

12. A conducting rod being pulled from rest to move with a constant acceleration in a uniform magnetic field on two horizontal tracks of negligible resistance is shown in the diagram below.

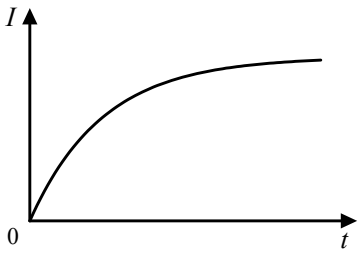


The tracks are connected to a resistor R at both of its ends. Which graph shows the variation of the induced current I with time t for the conducting rod?

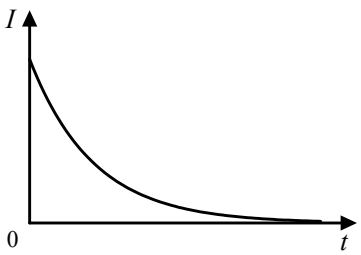
A



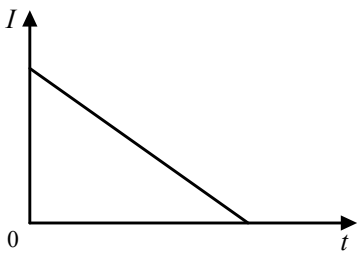
B



C



D

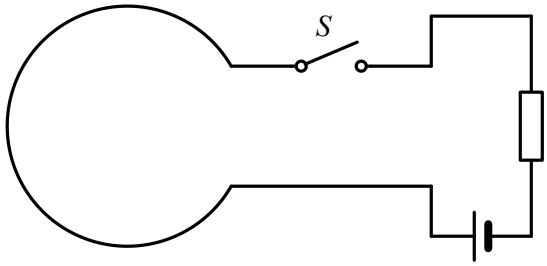


Answer: A

$I = \frac{Blv}{R}$ with $v = at \Longrightarrow I = \frac{Bla}{R}t$: a straight line through the origin (A)

Explanation: The rod is pulled so that its acceleration stays constant; some answer books give C.

13. A circular conducting loop is connected to a battery and a switch S as shown in the diagram below.



Immediately after S is closed, the current through the loop changes at a rate of 15 A s^{-1}, and the e.m.f. induced in the loop has a magnitude of 5.0 V. What is the self-inductance of the loop?

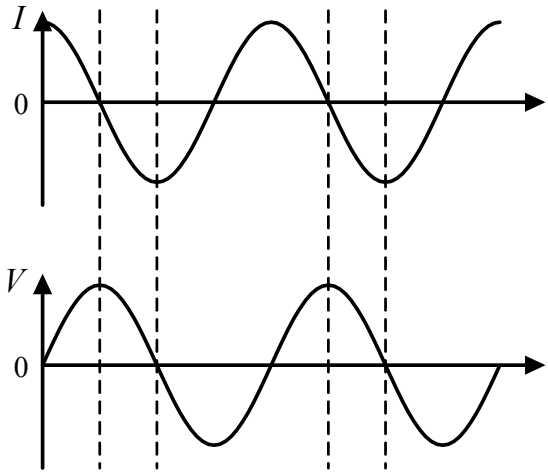
A 3.0 H B 5.8×10^{-1} H C 3.3×10^{-1} H D 4.4×10^{-2} H

Answer: C

$$L = \frac{\varepsilon}{\mathrm{d}I/\mathrm{d}t} = \frac{5.0}{15} = 0.33 \text{ H}$$

Explanation: Self-inductance of a single loop.

14. The variation of current I and voltage V with time t for a component X in an a.c. circuit is shown in the graphs below.



What is the component X?

A A diode B A resistor C A capacitor D An inductor

Answer: C

The current reaches its maximum a quarter cycle before the voltage: the current leads the voltage by 90°, so X is a capacitor.

Explanation: In an inductor the current would lag the voltage by 90°.

15. A capacitor is connected to an a.c. source with a peak voltage of 48 V and a frequency of 90 Hz. If the capacitance is 3.7 μF, what is the peak current in the circuit?

A 0.00 mA B 71.0 mA C 100 mA D 140 mA

Answer: C

$$I_0 = V_0(2\pi fC) = (48)(2\pi)(90)(3.7 \times 10^{-6}) = 0.100 \text{ A} = 100 \text{ mA}$$

Explanation: $X_C = 1/(2\pi fC) = 478\ \Omega$.

Section B [15 marks]

16. (a) Derive an expression for the capacitance of a parallel plate capacitor in terms of the overlapping area A, the separation distance d, the permittivity of vacuum ε_0 and the relative permittivity ε_r of the material filling up the gap between the plates. [4]

Answer

By Gauss's law the field between the plates (in vacuum) is $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$ 1

With the dielectric filling the gap, $E = \frac{Q}{\varepsilon_r \varepsilon_0 A}$ 1

$V = Ed = \frac{Qd}{\varepsilon_r \varepsilon_0 A}$ 1

$C = \frac{Q}{V} = \frac{\varepsilon_r \varepsilon_0 A}{d}$ 1

4

Explanation: The field between large parallel plates is uniform.

(b) Explain briefly, in microscopic view, how the presence of a dielectric material between a parallel plate capacitor can increase its capacitance. [3]

Answer

The field polarises the molecules of the dielectric (dipoles align or are induced) 1

Induced surface charges on the dielectric produce a field opposite to the applied field, so the net field and p.d. between the plates decrease 1

For the same charge, $C = \frac{Q}{V}$ increases 1

3

Total for question **7**

Explanation: C increases by the factor ε_r.

17. (a) State **two** factors which affect the resistivity of a conducting material. [2]

Answer

The number of free electrons per unit volume (the type of material)	1
The temperature	1
	2

Explanation: Resistivity does not depend on the length or area of the sample.

(b)

(i) Show that the conductivity σ of a conducting material can be expressed as $\sigma = \frac{ne^2\tau}{m}$, where n is the number of free electrons, e is the charge of an electron, m is the mass of an electron and τ is the mean time between collisions of free electrons and positive ions. [4]

Answer

An electron accelerates with $a = \frac{eE}{m}$ for a mean time τ between collisions	1
Drift velocity $v_d = \frac{eE\tau}{m}$	1
$J = nev_d = \frac{ne^2\tau}{m}E$	1
$\sigma = \frac{J}{E} = \frac{ne^2\tau}{m}$	1
	4

Explanation: $J = \sigma E$ is the microscopic form of Ohm's law.

(ii) Copper has a resistivity of 1.69×10^{-8} Ω m and a free electron density of 8.47×10^{28} m^{-3}. Estimate the mean time between the collisions of free electrons and positive ions in copper. [2]

Answer

$\tau = \frac{m}{ne^2\rho} = \frac{9.11 \times 10^{-31}}{(8.47 \times 10^{28})(1.60 \times 10^{-19})^2(1.69 \times 10^{-8})}$	1
$\tau = 2.49 \times 10^{-14}$ s	1
	2
Total for question	**8**

Explanation: $\sigma = 1/\rho$.

Section C [30 marks]

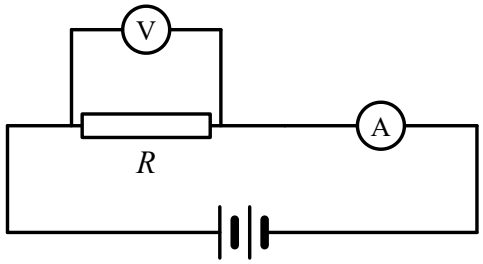
18. (a) Explain why a good moving coil ammeter has a low resistance while a good moving coil voltmeter has a high resistance. [4]

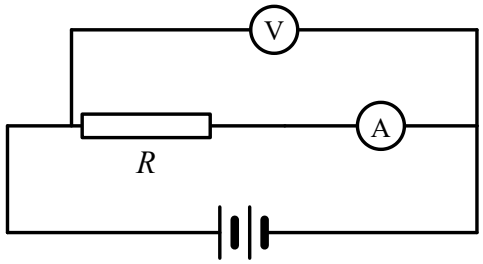
Answer

An ammeter is connected in series with the component	1
A low resistance hardly changes the current it is measuring	1
A voltmeter is connected in parallel with the component	1
A high resistance draws very little current, so it hardly changes the p.d. it is measuring	1
	4

Explanation: An ideal ammeter has zero resistance; an ideal voltmeter infinite resistance.

(b) The resistance of a resistor R can be determined using a moving coil ammeter and voltmeter as shown in the circuit P or circuit Q below.





(i) Explain why the value of the resistance R determined is not accurate in each case. [2]

Answer

Circuit P: the ammeter reads the current through R plus the current through the voltmeter, so V/I is too small	1
Circuit Q: the voltmeter reads the p.d. across R plus the p.d. across the ammeter, so V/I is too large	1
	2

Explanation: Real meters have finite resistances.

(ii) Explain which circuit is a better way of determining a small resistance. [2]

Answer

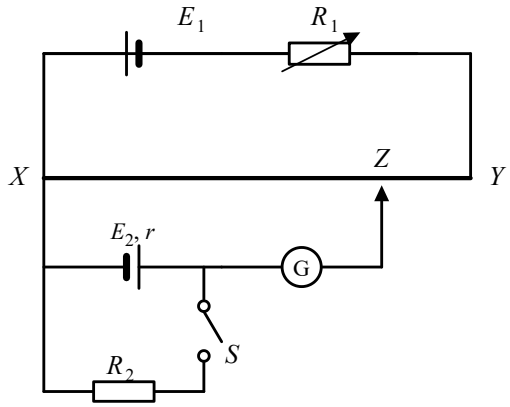
Circuit P — 1

For a small R, the voltmeter (high resistance) current is negligible compared with the current in R, whereas in Q the ammeter resistance would be comparable with R — 1

2

Explanation: Circuit Q is better for a large resistance.

(c) A potentiometer which is used to determine the internal resistance and the e.m.f. of a dry cell E_2 is shown in the diagram below.



The e.m.f. of the driver cell E_1 is 2.0 V with negligible internal resistance. The variable resistor R_1 is set at 1.0 Ω and the slide wire XY which is 100 cm long and has a resistance of 3.0 Ω. The resistor R_2 has a resistance of 5.0 Ω. When the switch S is opened, the balance length XZ is 90.0 cm, and when the switch S is closed the balance length changes to 80.0 cm.

(i) Determine the potential difference across the slide wire XY. [2]

Answer

$I = \dfrac{2.0}{1.0 + 3.0} = 0.50$ A — 1

$V_{XY} = (0.50)(3.0) = 1.5$ V — 1

2

Explanation: R_1 and the wire are in series with the driver cell.

(ii) Determine the e.m.f. of the cell E_2. [2]

Answer

With S open, no current is drawn from E_2: $E_2 = 1.5 \times \frac{90.0}{100}$ 1

$E_2 = 1.35$ V 1

2

Explanation: The potentiometer measures the e.m.f. when no current flows in the cell.

(iii) Calculate the internal resistance r of the cell E_2. [3]

Answer

S closed: terminal p.d. $V = 1.5 \times \frac{80.0}{100} = 1.20$ V 1

$I = \frac{V}{R_2} = \frac{1.20}{5.0} = 0.24$ A 1

$r = \frac{E_2 - V}{I} = \frac{1.35 - 1.20}{0.24} = 0.625\ \Omega$ 1

3

Total for question **15**

Explanation: Equivalently $r = R_2(l_1/l_2 - 1)$.

19. (a)

(i) State the laws of electromagnetic induction. [2]

Answer

Faraday: the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage 1

Lenz: the induced current flows in a direction that opposes the change producing it 1

2

Explanation: $\varepsilon = -\mathrm{d}(N\Phi)/\mathrm{d}t$.

(ii) Derive an expression for the induced e.m.f. in a piece of steel rod of length l which is moving with a velocity v across a uniform magnetic field with a flux density B. [3]

Answer

In time Δt the rod moves $v\Delta t$ and sweeps an area $\Delta A = lv\Delta t$ 1

Flux cut $\Delta\Phi = B\Delta A = Blv\Delta t$ 1

$\varepsilon = \frac{\Delta\Phi}{\Delta t} = Blv$ 1

3

Explanation: For **v** perpendicular to **B** and to the rod.

(b) A pair of smooth conducting rails, in a uniform magnetic field, is inclined at 20° to the ground. A 1.20 m long steel rod of mass 1.0 kg is allowed to slide down from rest on the rails. Assume that the field of flux density 0.53 T is perpendicular to the ground.

(i) Determine the magnitude of the induced e.m.f. in the rod at an instant when its speed is 5.00 m s^{-1}. [2]

Answer

The field is vertical, so only the horizontal component of the velocity, v cos 20°, cuts the field: $\varepsilon = Blv \cos 20°$	1
$\varepsilon = (0.53)(1.20)(5.00) \cos 20° = 2.99$ V	1
	2

Explanation: The velocity is along the inclined rails.

(ii) One end of both rails are joined with a wire of resistance 2.5 Ω forming a closed circuit. If the rod and rails have negligible resistance, calculate the terminal velocity of the rod. [5]

Answer

$I = \dfrac{Blv \cos 20°}{R}$	1
The magnetic force BIl on the rod is horizontal; its component up the incline is BIl cos 20°	1
At terminal velocity: $BIl \cos 20° = mg \sin 20°$	1
$\dfrac{B^2 l^2 v \cos^2 20°}{R} = mg \sin 20° \Rightarrow v = \dfrac{(1.0)(9.81)(2.5) \sin 20°}{(0.53)^2(1.20)^2 \cos^2 20°}$	1
$v = 23.5$ m s^{-1}	1
	5

Explanation: The rails are smooth, so there is no friction.

(iii) Describe the transformation of energy in (b)(ii). [3]

Answer

At terminal velocity the kinetic energy of the rod is constant	1
The loss in gravitational potential energy is converted into electrical energy (induced current)	1
which is dissipated as heat (thermal energy) in the resistor	1
	3
Total for question	**15**

Explanation: Rate of loss of GPE = $mgv \sin 20° = I^2R$.

20. (a) Distinguish between a direct current and an alternating current. [2]

Answer

A direct current flows in one direction only (its magnitude may be constant) 1

An alternating current reverses its direction periodically (e.g. sinusoidally) 1

2

Explanation: A.c. averages to zero over a cycle.

(b) The output of an a.c. supply is given by the expression $V = 30 \sin 6000t$, where V is in volts and t in seconds. Calculate

(i) the maximum voltage, [1]

Answer

$V_0 = 30$ V 1

1

Explanation: The amplitude in $V = V_0 \sin \omega t$.

(ii) the r.m.s. voltage, [2]

Answer

$V_{rms} = \frac{30}{\sqrt{2}}$ 1

$V_{rms} = 21.2$ V 1

2

Explanation: For a sinusoidal voltage.

(iii) the frequency of the current supply. [2]

Answer

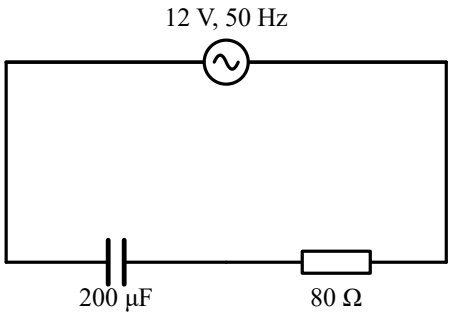
$\omega = 6000 \text{ rad s}^{-1} \Rightarrow f = \frac{6000}{2\pi}$ 1

$f = 955$ Hz 1

2

Explanation: $\omega = 2\pi f$.

(c) A capacitor of capacitance 200 μF and a resistor of resistance 80 Ω which are connected in series to an a.c. supply of r.m.s. voltage V_S of 12 V with a frequency of 50 Hz are as shown in the circuit below.



(i) Calculate the r.m.s. current of the circuit. [4]

Answer

$X_C = \dfrac{1}{2\pi(50)(200 \times 10^{-6})} = 15.9\ \Omega$ 1

$Z = \sqrt{(80^2 + 15.9^2)}$ 1

$Z = 81.6\ \Omega$ 1

$I_{rms} = \dfrac{12}{81.6} = 0.147\ \text{A}$ 1

4

Explanation: Resistance and reactance add as phasors.

(ii) Calculate the r.m.s. voltage across the resistor V_R and the r.m.s. voltage across the capacitor V_C. [2]

Answer

$V_R = (0.147)(80) = 11.8\ \text{V}$ 1

$V_C = (0.147)(15.9) = 2.34\ \text{V}$ 1

2

Explanation: $11.8 + 2.34 \neq 12$: the voltages are not in phase.

(iii) State the relationship between V_R, V_C and V_S. Give a reason for your answer. [2]

Answer

$V_S^2 = V_R^2 + V_C^2$ 1

because V_C lags V_R (and the current) by 90°, so they add as perpendicular phasors 1

2

Total for question **15**

Explanation: Check: $\sqrt{(11.8^2 + 2.34^2)} = 12$ V.